

Predictive Pricing & Reinsurance Analytics

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Abstract

This report outlines the development of a data pipeline and predictive pricing model for Property & Casualty (P&C) reinsurance. The project processes 423,334 policy records using DuckDB. It compares a traditional Generalized Linear Model (GLM) baseline with an XG-Boost architecture to calculate the portfolio's pure premium, classify Excess of Loss (XoL) tail-risk, and identify areas of underwriting leakage.

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1 Data Pipeline and Financial Metrics

The project integrates real-world exposure data with stochastic simulations to replicate a standard Reinsurance Finance dataset.

- 1. Exposure & Pricing (freMTPL2):** The core motor insurance portfolio, containing demographic, geographic, and vehicle characteristics.
- 2. Claims Reserving Simulator:** A stochastic layer that generates Reporting Delays and Payment Delays to classify claims into Paid, Case Reserves, and IBNR (Incurred But Not Reported).
- 3. Reinsurance & Expense Layer:** A synthetic financial layer that applies operational expenses (OPEX), reinsurance commissions (15%), and a 50% Quota Share treaty structure.

1.1 Financial Metrics Definition

The pipeline calculates standard P&C Key Performance Indicators (KPIs). SQL NULLIF was used to prevent zero-division errors during aggregation.

Table 1: Financial Metrics Glossary

Metric	Description	Formula
Loss Ratio (LR)	Technical underwriting quality. Out-flow for claims vs. collected premiums.	$\frac{\text{Incurred Losses}}{\text{Earned Premiums}}$
Expense Ratio (ER)	Operational efficiency and acquisition costs.	$\frac{\text{Underwriting Expenses} + \text{OPEX}}{\text{Earned Premiums}}$
Combined Ratio (CR)	Primary profitability metric. Values > 100% indicate technical losses.	$LR + ER$
IBNR Reserve	Capital set aside for claims that occurred but remain unreported.	Ultimate – Reported
Claim Frequency	Rate of claim occurrences relative to time on risk.	$\frac{\text{Number of Claims}}{\text{Exposure}}$

1.2 Portfolio Baseline Results

The foundational data layer processed the portfolio and generated the following technical baseline.

Table 2: Global Portfolio Financial Baseline

Portfolio Metric	Aggregated Value
Total Processed Policies	423,334
Average Portfolio Loss Ratio (incl. IBNR)	42.77%
Average Portfolio Expense Ratio	8.01%
Portfolio Combined Ratio (CR)	50.78%

1.3 Database Structure (DuckDB)

The final database is structured into a normalized schema to support direct BI querying. Below are samples from the generated tables.

1.3.1 Table: Dim_Policies (Dimension)

IDpol	Exposure	Region	VehPower	DrivAge	BonusMalus	Claim Freq	Und. Date	Und. Month
139	0.75	R82	5	50	50	1.3333	2024-03-15	2024-03
190	1.00	R22	6	28	65	0.0000	2025-08-21	2025-08
214	0.50	R11	9	22	100	2.0000	2026-01-05	2026-01

1.3.2 Table: fact_claims (Reserving Fact Table)

IDpol	TotalClaim	Rep. Delay	Tot. Delay	Status	Paid Amt	Case Rsv	IBNR Rsv	Ultimate Loss
139	1,250.00	4	28	Paid Closed	1,250.00	0.00	0.00	1,250.00
214	45,000.00	12	315	Reported Open	0.00	45,000.00	0.00	45,000.00
899	8,500.00	114	518	IBNR	0.00	0.00	8,500.00	8,500.00

1.3.3 Table: Fact_Financials (Financial Reporting Fact Table)

IDpol	Earned Prem.	Paid Loss	IBNR Rsv	Ultimate Loss	OPEX	Loss Ratio	Comb. Ratio	Ceded Prem.
139	337.50	1,250.00	0.00	1,250.00	69.37	370.37%	390.92%	168.75
190	452.50	0.00	0.00	0.00	30.00	0.00%	6.63%	226.25
899	600.00	0.00	8,500.00	8,500.00	227.50	1416.67%	1454.58%	300.00

2 Linear Modeling Baseline (GLMs)

Before applying machine learning, a Generalized Linear Model (GLM) baseline was established. This provides a standard multiplicative tariff benchmark.

2.1 Mathematical Framework

Frequency Model (Poisson): Models the claim counts. $\log(Exposure)$ is used as an offset to account for variable policy durations.

$$\log(E[\text{Count}]) = \log(\text{Exposure}) + \beta_0 + \sum_{i=1}^n \beta_i X_i \quad (1)$$

Severity Model (Gamma): Models the strictly positive cost of claims.

$$\log(E[\text{Severity}]) = \beta_0 + \sum_{i=1}^n \beta_i X_i \quad (2)$$

2.2 GLM Results

The `statsmodels` output yields the following feature significance. The exponential of the coefficients (e^β) represents the multiplier applied to the base rate.

Table 3: Frequency Model (Poisson GLM)

Feature	Coefficient (β)	Multiplier (e^β)	P-Value	Significance
Intercept	-3.6655	0.025	0.000	***
C(Area) [T.'C']	0.0538	1.055	0.006	***
C(Area) [T.'F']	0.2809	1.324	0.000	***
VehPower	0.0176	1.017	0.000	***
VehAge	-0.0579	0.943	0.000	***
DrivAge	0.0083	1.008	0.000	***
BonusMalus	0.0225	1.022	0.000	***

Table 4: Severity Model (Gamma GLM)

Feature	Coefficient (β)	Multiplier (e^β)	P-Value	Significance
Intercept	6.0007	403.71	0.000	***
C(Area)[T.'F']	-0.1981	0.820	0.005	***
VehAge	0.0377	1.038	0.000	***
BonusMalus	0.0070	1.007	0.000	***

2.3 Model Observations & Limitations

- **Geographic Impact:** Moving from Area A (baseline) to Area F increases claim frequency by 32.4%.
- **Vehicle Age Effect:** Older vehicles have lower claim frequencies ($\beta = -0.0579$) but higher severities when claims do occur ($\beta = +0.0377$).
- **Limitations of GLM:** The model forces a linear relationship on variables like Driver Age (assigning a static +0.8% penalty per year). It fails to capture the known U-shaped risk curve of driver age. Furthermore, the Poisson distribution assumes equidispersion (Mean = Variance), which limits accuracy on fat-tailed distributions.

3 Non-Linear Pricing and Tail-Risk (XGBoost)

To address the limitations of the GLM, an XGBoost architecture was implemented to capture non-linear relationships and feature interactions.

3.1 Pure Premium Calculation

- **Frequency Engine** (`count:poisson`): Predicts claim counts, using the natural logarithm of exposure as the base margin.
- **Severity Engine** (`reg:gamma`): Predicts claim costs. The training set was restricted to policies where *UltimateLosses* > 0 to satisfy the constraints of the Gamma distribution.

The Expected Pure Premium for each policy is computed as:

$$E[\text{Loss}] = E[\text{Frequency}] \times E[\text{Severity}] \quad (3)$$

3.2 Excess of Loss (XoL) Tail-Risk Classifier

To support reinsurance treaty structuring, we defined a Retention Threshold at the 95th percentile of historical claim costs, which equates to **€4,812.00**.

An `XGBClassifier` (`binary:logistic`) was trained to predict the probability of a claim breaching this threshold. The `scale_pos_weight` parameter was applied to balance the dataset and prioritize the detection of these minority-class severe events.

4 Model Evaluation and Underwriting Leakage

The final step translates the model’s outputs into actionable pricing insights.

4.1 Feature Importance

To evaluate the impact of demographic factors on the model’s predictions, we utilized Permutation Importance. This method calculates the drop in model accuracy (Negative Mean Poisson Deviance) when a single feature is randomly shuffled.

$$I_j = s - s_{j,\text{perm}} \quad (4)$$

This approach provides a robust, model-agnostic ranking of risk drivers without the heavy computational overhead of other explainability frameworks.

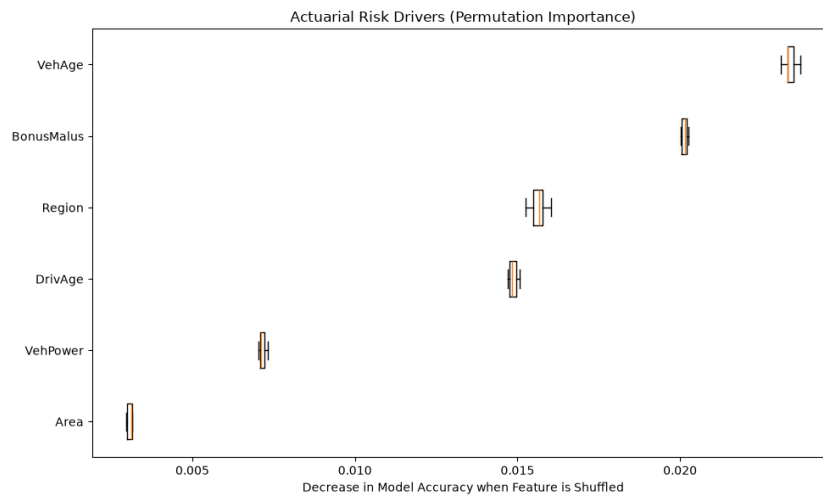


Figure 1: Feature Importance via Permutation (Decrease in accuracy).

4.2 Underwriting Leakage Analysis

We profiled the portfolio to identify structural mispricing. The Technical Loss Ratio (TLR) compares the model’s expected pure premium to the commercial premium charged:

$$\text{Technical LR} = \left(\frac{E[\text{Loss}]_{\text{XGBoost}}}{\text{Commercial Earned Premium}} \right) \times 100 \quad (5)$$

While the overall portfolio is profitable, filtering for policies with a TLR > 150% identified 23,291 underpriced policies. Grouping these via SQL revealed specific demographic and geographic clusters that consistently generate losses.

Table 5: Top 10 Underwriting Leakage Clusters (TLR > 150%)

Region	VehPower	Average TLR	Policy Count	Expected Losses (EUR)
R21	6	475.80%	66	37,594.37
R21	7	455.24%	82	50,721.08
R24	7	415.97%	20	190,836.21
R21	4	369.73%	72	32,240.92
R11	5	327.67%	958	418,755.76
R74	5	306.76%	51	16,765.69
R94	4	304.25%	144	49,698.56
R82	4	296.98%	721	303,692.32
R73	5	294.48%	241	97,624.34
R24	4	292.96%	187	99,522.75

4.3 Business Implications

- **Region R21 Discrepancy:** High-power vehicles (Power 6 and 7) in Region R21 exhibit TLRs over 450%, indicating that expected losses are roughly 4.5 times the collected premium.
- **Region R11 Volume Risk:** Region R11 (Vehicle Power 5) presents the highest absolute financial exposure due to volume. The cluster contains 958 underpriced policies, carrying an expected loss exceeding €418,000.
- **Risk Interaction Validation:** The highest TLRs consistently correlate with higher vehicle power interacting with specific regions. The XGBoost model successfully maps these interactions, highlighting risks that the baseline GLM smoothed over.

5 Conclusion

This project illustrates a complete data and modeling pipeline for P&C reinsurance. By leveraging DuckDB for data aggregation and transitioning from standard GLMs to XGBoost, the analysis accurately prices individual policies, establishes XoL thresholds, and successfully isolates structural underwriting leakage to inform targeted tariff revisions.